XLVII National Mathematics Olympiad: 4th round, 16–17 May 1998

Problem 1. Let n be a natural number. Find the least natural number k for which there exist k sequences of 0's and 1's of length $2n+2$ with the following property: any sequence of 0's and 1's of length $2n+2$ coincides in at least $n+2$ positions with some of these k sequences.

Problem 2. The polynomials $P_n(x,y)$, $n = 1, 2, \ldots$ are defined by

$$P_1(x,y) = 1, P_{n+1}(x,y) = (x+y-1)(y+1)P_n(x,y+2)+(y-y^2)P_n(x,y).$$

Prove that $P_n(x,y) = P_n(y,x)$ for all x, y and n.

Problem 3. On the sides of a non-obtuse triangle ABC a square, a regular n-gon and a regular m-gon $(n, m > 5)$ are constructed externally, so that their centres are vertices of a regular triangle. Prove that $m = n = 6$ and find the angles of ABC.

Problem 4. Let $a_1, a_2, \ldots, a_n$ be real numbers, not all of them zero. Prove that the equation

$$\sqrt{1+a_1x} + \sqrt{1+a_2x} + \ldots + \sqrt{1+a_nx} = n$$

has at most one nonzero real root.

Problem 5. Let m and n be natural numbers such that $A = ((m+3)^n + 1)/3m$ is integer. Prove that A is an odd integer number.

Problem 6. The sides and the diagonals of a regular n-gon X are coloured in k colours so that:

(i) segment AB is coloured in colour a or there is a vertex C such that AC and BC are coloured in colour a;

(ii) the sides of any triangle with vertices among the vertices of X are coloured in at most two colours.

Prove that $k \leq 2$.

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